

What is the transitive closure?

Given a set A , and a relation $R \subseteq A \times A$, we define $R^+ \subseteq A \times A$, and $R^* \subseteq A \times A$ as

- $R^*(a, a') \triangleq a = a' \vee \exists a'' \in A. [R(a, a'') \wedge R^*(a'', a')]$
- $R^+(a, a') \triangleq R(a, a') \vee \exists a'' \in A. [R(a, a'') \wedge R^+(a'', a')]$

R^* and R^+ are inductively defined, and therefore give rise to inductive principles.

$$\begin{array}{c} R^*-A \quad \frac{}{R^*(a, a)} \quad a \in A \\ R^*-B \quad \frac{R(a, a'') \quad R^*(a'', a')}{R^*(a, a')} \\ R^+-A \quad \frac{R(a, a')}{R^+(a, a')} \\ R^+-B \quad \frac{R(a, a'') \quad R^+(a'', a')}{R^+(a, a')} \end{array}$$